

A Primitive Four-versus-Four Burnside Relation from a $W(E_6)$ Lambda-Square Shadow

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Abstract

Equal permutation characters can hide relations among nonconjugate finite sets. We expose one such relation in the degree-320 Hénon $W(E_6)$ -Gassmann construction. Starting from the complete exterior- and symmetric-square atlas of the two C61 permutation sets, we rebind the ambient group, enumerate its 25 conjugacy classes, and recompute the fixed-coset character matrix on the 16 ambient-conjugacy subgroup types. The resulting 25×16 integer matrix has rank 13. Its exterior-square difference is the primitive four-versus-four relation

$$[G/S_2] + [G/S_3] + [G/S_5] + [G/S_6] = [G/S_{11}] + [G/S_{12}] + [G/S_{13}] + [G/S_{14}],$$

where the eight core-free types have degrees 480, 480, 4320, 4320 on each side. A support-restricted rank-7 certificate proves that no proper subset supports a further relation. We also locate the original C61 relation and an inherited C60 collision inside the full three-dimensional restricted kernel. The result is finite-group/Burnside evidence only; no arithmetic resolvent, local-field, bad-Euler-factor, or root-number claim is made.

1 Introduction

Gassmann equivalence is deliberately subtle: two transitive G -sets can have the same rational permutation character even when their stabilizers are not conjugate. The first two degree-two lambda shadows of the C61 $W(E_6)$ pair make this tension visible. C62 computed the complete exterior-square and symmetric-square orbit atlases, but left open a structural question: which relation in the linearization kernel is actually being seen by the exterior-square atlas?

This paper answers that question on a precisely bounded submodule. We take the 16 subgroup-conjugacy types recorded by C62, evaluate their transitive permutation characters on all 25 conjugacy classes of $W(E_6)$, and perform an exact rational rank computation. The exterior-square difference is not merely a vanishing character vector. It is a support-minimal four-versus-four Burnside relation among eight nonconjugate fixed-field types.

Our contributions are:

1. We give an independent, source-bound reconstruction of the 25-class $W(E_6)$ character matrix on the 16 C62 types and certify rank 13 and nullity 3.
2. We identify the primitive relation R_4 exposed by the exterior square and prove support minimality by a rank-7 restriction, rather than by subgroup order or a table position.
3. We place the relation next to the original C61 Gassmann direction and the symmetric-square relation, while explicitly treating the $S_9 - S_{10}$ collision as inherited C60 material rather than a new claim.

All statements are scoped by `NO_BAD_EULER_OR_ROOT_NUMBER`. In particular, the word “field” below means the core-free fixed-field type supplied by the C62 finite-group atlas; we do not infer local arithmetic data from its degree or normalizer.

2 Relation to prior work

The linearization map from finite G -sets to permutation characters is a classical Burnside-ring construction [1]. Gassmann's criterion and its number-field interpretation explain why equality of these characters can coexist with nonconjugate stabilizers [2, 3]. C63 uses these general mechanisms only as background. Its claim is the bounded, instance-specific computation on the 16 subgroup types produced by C62: the lambda-square difference selects an eight-column support whose restricted kernel is one-dimensional. We do not claim a new general theorem about Burnside rings or Gassmann equivalence.

3 Finite-group setup

Let $G = W(E_6)$, with $|G| = 51840$, and let H_+, H_- be the two order-162 stabilizers in the released C61 source contract. Put $X_{\pm} = G/H_{\pm}$, so $|X_{\pm}| = 320$. C62 enumerated the G -orbits on 2-subsets and size-two multisets of each $X_{\pm}$, and grouped the complete stabilizer element sets by ambient conjugacy. We denote the resulting 16 types by $S_1, \dots, S_{16}$, and write

$$Y_i = [G/S_i]$$

for the corresponding transitive basis symbols.

For $g \in G$, let $C(g)$ be its conjugacy class and let $c(g) = |C_G(g)|$. The fixed-coset formula used in the producer is

$$\chi_{G/S}(g) = \frac{c(g) |S \cap C(g)|}{|S|}. \quad (1)$$

It follows by counting pairs (x, s) with $x^{-1}gx = s$. Formula (1) uses complete subgroup element sets, so it does not identify types using only $|S|$.

The C62 lambda identities explain why a relation must occur:

$$\chi_{\Lambda^2 X}(g) = \frac{\chi_X(g)^2 - \chi_X(g^2)}{2}, \quad \chi_{\text{Sym}^2 X}(g) = \frac{\chi_X(g)^2 + \chi_X(g^2)}{2}.$$

Since C61 has $\chi_{X_+} = \chi_{X_-}$, both lambda differences linearize to zero. C63 determines their exact support in the 16-type basis.

4 Main result

Let M be the matrix with one row for each of the 25 conjugacy classes of G , one column for each Y_i , and entries given by (1).

Theorem 1 (Restricted kernel and primitive lambda relation). *The matrix M has*

$$\text{rank}_{\mathbb{Q}} M = 13, \quad \dim_{\mathbb{Q}} \ker M = 3.$$

A basis is

$$\begin{aligned} z_1 &= Y_{10} - Y_9, \\ z_2 &= -Y_2 - Y_3 - Y_5 - Y_6 + Y_{11} + Y_{12} + Y_{13} + Y_{14}, \\ z_3 &= Y_{16} - Y_{15}. \end{aligned}$$

The C61 difference is $r = Y_{15} - Y_{16} = -z_3$. The C62 differences are

$$q_\Lambda = Y_2 + Y_3 + Y_5 + Y_6 - Y_{11} - Y_{12} - Y_{13} - Y_{14} = -z_2, \quad q_{\text{Sym}} = q_\Lambda + r.$$

On the eight-term support of q_Λ , the restricted matrix has rank 7. Hence its rational kernel is one-dimensional and q_Λ has no nonzero proper-support subrelation.

The theorem is intentionally a statement about the 16-type submodule, not a classification of the full Burnside ring of G . The vector z_1 is an inherited C60 order-4 collision and is included to prevent a false novelty claim.

5 The four-versus-four support

The eight terms of q_Λ split into two degree blocks. Every listed subgroup is core-free; the order and normalizer data come directly from the C62 element-set atlas.

side	type labels				degrees
plus	S_2	S_3	S_5	S_6	480, 480, 4320, 4320
minus	S_{11}	S_{12}	S_{13}	S_{14}	480, 480, 4320, 4320

The order-108 pair (S_2, S_3) and (S_{11}, S_{12}) contributes degree 480 fields; the order-12 pair (S_5, S_6) and (S_{13}, S_{14}) contributes degree 4320 fields. The four plus types and four minus types are distinct ambient-conjugacy classes even when their subgroup orders agree. Their total degree is 9600 on each side.

To make “primitive” checkable, let M_{R_4} be the submatrix obtained by retaining only these eight columns. Exact elimination gives

$$\text{rank}_{\mathbb{Q}} M_{R_4} = 7, \quad \dim_{\mathbb{Q}} \ker M_{R_4} = 1.$$

Since all eight coefficients of q_Λ are nonzero, any vector in this one-dimensional kernel has the same support. This rules out a hidden two-term or smaller relation inside the C62 support.

6 Where the lambda shadows land

The diagonal multisets in $\text{Sym}^2 X_\pm$ form copies of $X_\pm$. Therefore the symmetric difference is forced to be the exterior difference plus the original C61 difference:

$$[\text{Sym}^2 X_+] - [\text{Sym}^2 X_-] = ([\Lambda^2 X_+] - [\Lambda^2 X_-]) + ([X_+] - [X_-]).$$

The producer recovers exactly $q_{\text{Sym}} = q_\Lambda + r$. This identity is useful as a semantic check: treating the symmetric atlas as an unrelated second relation would double-count the diagonal direction.

The remaining basis vector z_1 has equal character but comes from the degree-12960 types S_9, S_{10} . C60 already recorded that collision, so C63 uses it only as a control for the rank computation. The new content is the eight-term support exposed by the lambda-square shadow and the proof that it is primitive inside the recorded submodule.

7 Fixed-field and zeta interpretation

Every S_i in the C62 dictionary is core-free, and its fixed-field degree is $51840/|S_i|$. Thus the four-versus-four relation can be read as an equality of formal permutation characters attached to eight fixed fields in the common finite normal-closure model. When the corresponding Artin formalism is invoked, it gives the product identity

$$\prod_{i \in \{2,3,5,6\}} L(s, \mathbf{1}_{S_i}) = \prod_{i \in \{11,12,13,14\}} L(s, \mathbf{1}_{S_i}).$$

This sentence records only the character-level identity. We do not expand arithmetic resolvents, identify discriminants or maximal orders, classify local extensions, or separate bad Euler factors and root numbers.

8 Reproducibility

The producer reads the frozen C61 group bytes and the C62 atlas/dictionary bytes, verifies their SHA-256 digests, and reconstructs every subgroup and conjugacy class. The evidence matrix digest is `e912b0f37f69ac1e23cf432915aa4258818312f84fba776986876c7625a84a9b`. The commands are:

```
python3 -m py_compile code/*.py
python3 code/c63_kernel.py
python3 code/c63_kernel_checker.py
python3 code/c63_mutation_test.py
```

The hostile suite rejects mutations of source hashes, matrix entries, rank, kernel vectors, support rank, scope, and nonclaim flags.

9 Limitations and conclusion

C63 is scoped to a finite list of 16 core-free subgroup types inherited from the C62 atlas. It does not assert that the displayed three-dimensional kernel is the kernel of the full Burnside ring, nor does it make arithmetic local or analytic claims. Within that boundary, the lambda-square operation does more than reproduce a character equality: it exposes a support-minimal four-versus-four relation whose exact rank certificate distinguishes it from both the original C61 direction and the inherited C60 collision.

A Conjugacy-class audit

For reproducibility, the deterministic class-size vector used by the evidence is

$$(1, 36, 240, 270, 1440, 1620, 5184, 540, 2160, 1440, 6480, 3240, 4320, 540, 540, 45, 5184, 480, 1440, 4320, 4320, 5760, 720, 1440, 80).$$

It sums to 51840. The corresponding centralizer orders are obtained by dividing 51840 by each entry. The JSON evidence stores the representative permutation for every class, the full 25-by-16 matrix, and all relation vectors, so the displayed digest is not a compact substitute for the data.

References

- [1] William Burnside. *Theory of Groups of Finite Order*. Cambridge University Press, 1911.
- [2] Fritz Gassmann. Bemerkungen zu der vorstehenden arbeit des herrn t. petrick. *Mathematische Zeitschrift*, 25:665–675, 1926.
- [3] Robert Perlis. On the equation of dedekind zeta functions. *Journal of Number Theory*, 9(3): 342–360, 1977.